

Network Science Summer School

net-science.github.io



Universiteit Utrecht

Day program

9:00–12:00:

Introduction to network science: why networks matter and how to describe them

12:00–13:00:

Lunch

13:00–16:30:

Network representation: adjacency matrices and linear algebra

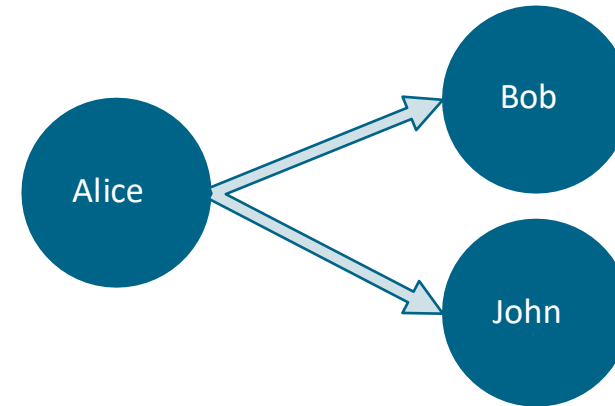
Who matters the most: centrality measures

01

Intro to linear algebra

Why? Many analyses on networks rely on matrix multiplication

Network representation



Adjacency list: (edgelist)

- Adv: It is dense: Only keeping edges
- Disadvantage: Hard to work with

Source	Target	Weight
Alice	Bob	1
Alice	John	1

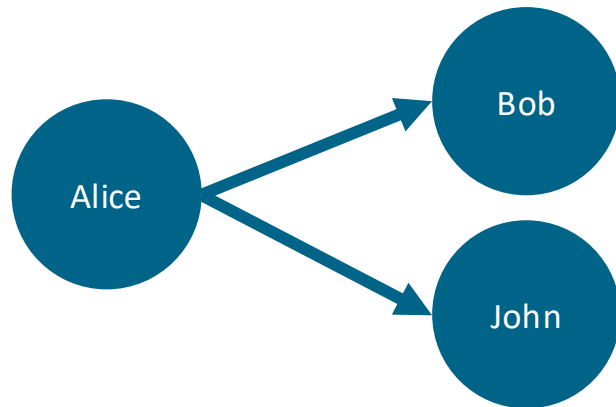
Adjacency matrix:

- Adv: Linear algebra is easy
- Disadvantage: It is sparse (mostly zeros). 1E6 nodes → 1 trillion options

Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	0	0	0
John	0	0	0

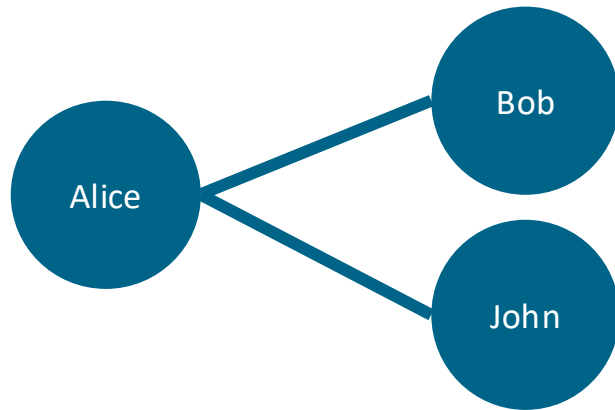
In computers → Sparse matrices: Best of both worlds

Directed networks



Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	0	0	0
John	0	0	0

Undirected networks



Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	1	0	0
John	1	0	0

Some terms

A =

Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	0	0	0
John	0	0	0

Diagonal

Trace = Sum of elements in the diagonal

Identity matrix (I) =

$I @ A = A$

	1	0	0
	0	1	0
	0	0	1

Transpose (A^T , A') =

(python) A.T

Target → ↓ Source	Alice	Bob	John
Alice	0	0	0
Bob	1	0	0
John	1	0	0

Symmetric matrix: $A = A.T$ (e.g. undirected network)

Python exercise notebook 2, ex.1

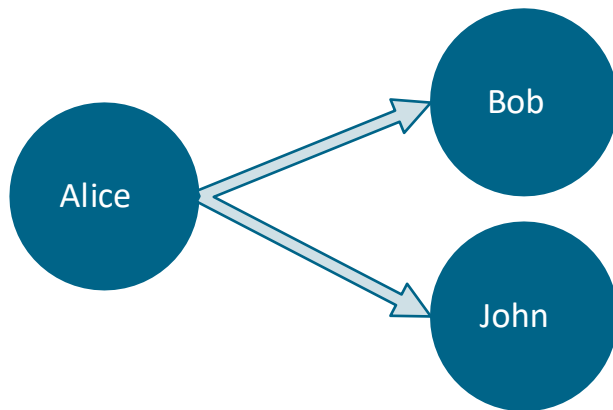
Python:

- Convert between formats
- Plot network with adjacency matrix

Transposing = reversing the edges

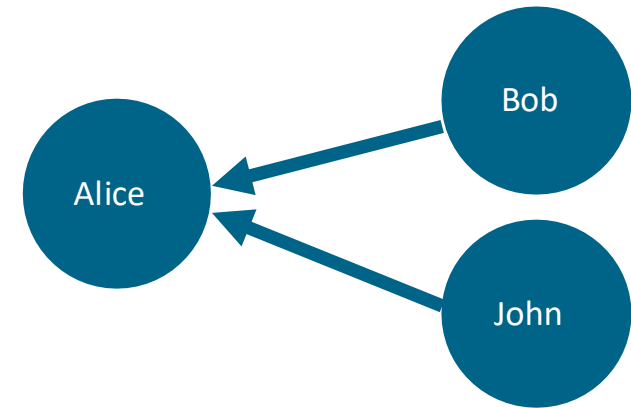
A =

Target → ↓ Source	Alice	Bob	John
Alice	0	1	1
Bob	0	0	0
John	0	0	0



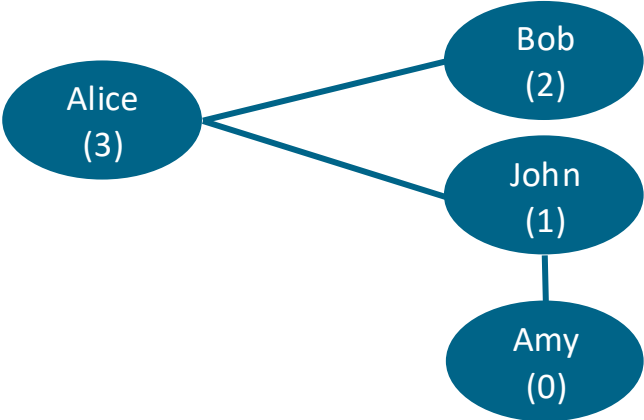
A.T =

Target → ↓ Source	Alice	Bob	John
Alice	0	0	0
Bob	1	0	0
John	1	0	0



Matrix multiplication: sum

[javier.science/marimo_intro_network_algebra](#) (Section 2)



Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Node	children
Alice	3
Bob	2
John	1
Amy	0

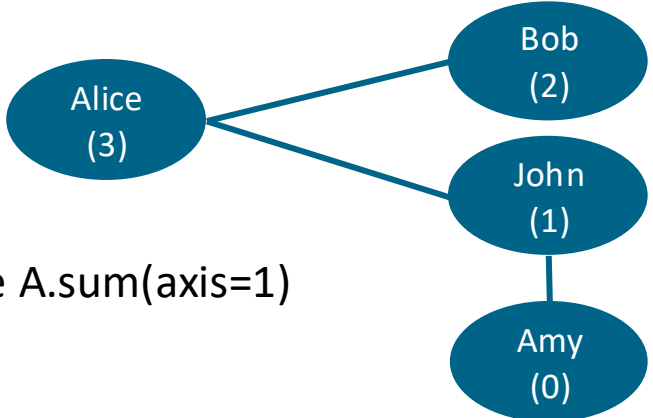
=

Node	kids
Alice	$0*3 + 1*2 + 1*1 + 0*0 = 3$
Bob	$1*3 + 0*2 + 0*1 + 0*0 = 3$
John	$1*3 + 0*2 + 0*1 + 1*0 = 3$
Amy	$0*3 + 0*2 + 1*1 + 0*0 = 1$

A @ M = SM

(N x N) @ (N x 1) = (N x 1)

Matrix multiplication: average



Divide by the degree. We get it by summing the adjacency elements column-wise $A.\text{sum}(\text{axis}=1)$

$$A @ M / A.\text{sum}(1)$$

$$(N \times N) @ (N \times 1) / (N \times 1) = (N \times 1) / (N \times 1) = (N \times 1)$$

Target → ↓ Origin	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Node	Kids
Alice	3
Bob	2
John	1
Amy	0

Node	Kids
Alice	3
Bob	3
John	3
Amy	1

=

=

Target → ↓ Source	Sum
Alice	2
Bob	1
John	2
Amy	1

Target → ↓ Source	Sum
Alice	2
Bob	1
John	2
Amy	1

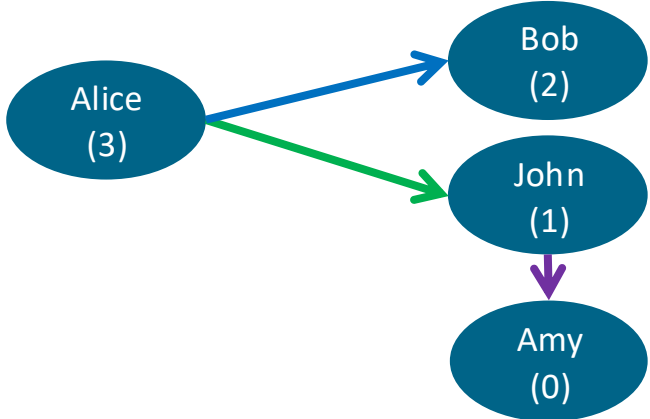
Node	Kids
Alice	1.5
Bob	3
John	1.5
Amy	1

Python exercise notebook 2, ex.2

Calculate the average number of children of your friends using matrix multiplication

Matrix multiplication: paths

Interpretation A: Presence of path between node i and j



A^2

=

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	1
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

From you → to your neighbors

From your neighbors → to their neighbors

You → the neig. of your neig.

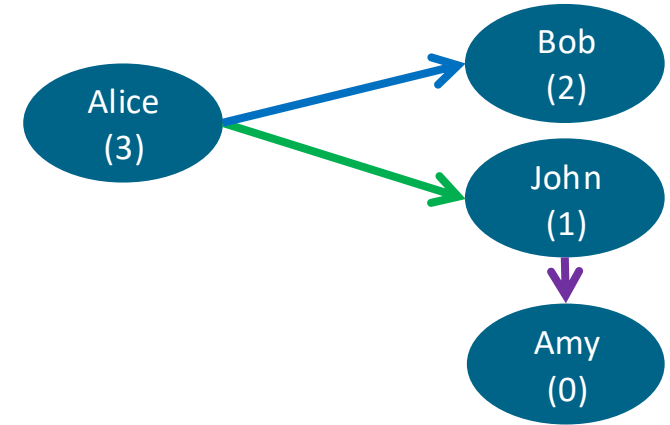
Matrix multiplication: paths

Interpretation A: Presence of path between node i and j

Interpretation A²: Number of paths between node i and j in two steps

Interpretation A³: Number of paths between node i and j in three steps

...



Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	1
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

$$\begin{aligned} & \text{Alice} \rightarrow \text{Alice (0)} * \text{Alice} \rightarrow \text{Amy (0)} \\ & + \text{Alice} \rightarrow \text{Bob (1)} * \text{Bob} \rightarrow \text{Amy (0)} \\ & + \text{Alice} \rightarrow \text{John (1)} * \text{John} \rightarrow \text{Amy (1)} \\ & + \text{Alice} \rightarrow \text{Alice (0)} * \text{Alice} \rightarrow \text{Amy (1)} \end{aligned}$$

Matrix multiplication: number of people reached in <3 steps

Number of paths in two or three steps from node i to node j: $N = A + A^2 + A^3$

We need to remove duplicate paths: $N = N > 0$

We need to remove paths from us to ourselves $N.setdiag(0)$

Python exercise notebook 2, ex.3.1

Matrix multiplication: number of triangles

We are interested in the diagonal of A^3

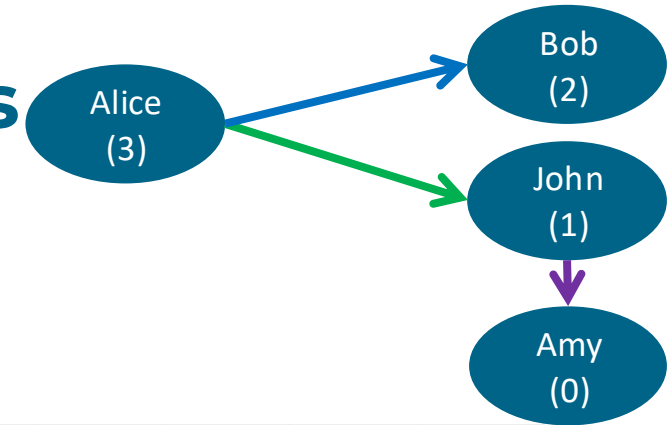
A triangle is a path from node i to node i .

The diagonal contains the number of triangles in which node i appears.

Undirected network? Divide the triangles by two (two directions)

Counting the total number of triangles? Divide the trace by 3 (each triangle has 3 members)

Matrix multiplication: number of triangles



The diagonal of A^3 counts the triangles through each node

A^2

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	1
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	0
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

Alice → Alice in two steps * Alice → Alice (0)
 Alice → Bob in two steps * Bob → Alice (0)
 Alice → John in two steps * John → Alice (0)
 Alice → Amy in two steps * Amy → Alice (0)

Diagonal of A^3

Alice → x_1 * $x_1 \rightarrow x_1$ * $x_1 \rightarrow$ Alice +
 Alice → x_1 * $x_1 \rightarrow x_2$ * $x_2 \rightarrow$ Alice +
 ...

Python exercise notebook 2, ex.3.2—3.3

02

Centrality measures

Key concept 3 · Who are the key actors in the network?

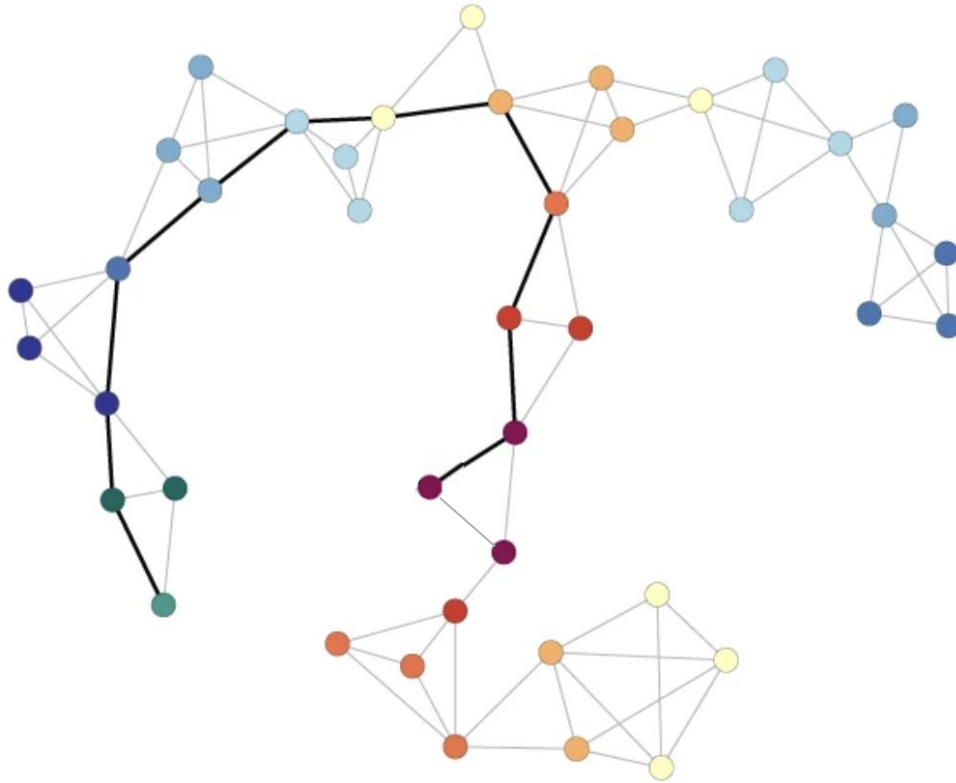
<https://aksakalli.github.io/2017/07/17/network-centrality-measures-and-their-visualization.html>

https://javier.science/marimo_intro_networks/

Motivating examples

How to stop the spread of diseases?

c



Important nodes: bridges

How to sort Google results?

PageRank counts the **quality** and **quantity** of backlinks to assess the importance of a page.



<https://www.leannewong.co/google-pagerank/>

Important nodes: those linked by important nodes

Centrality

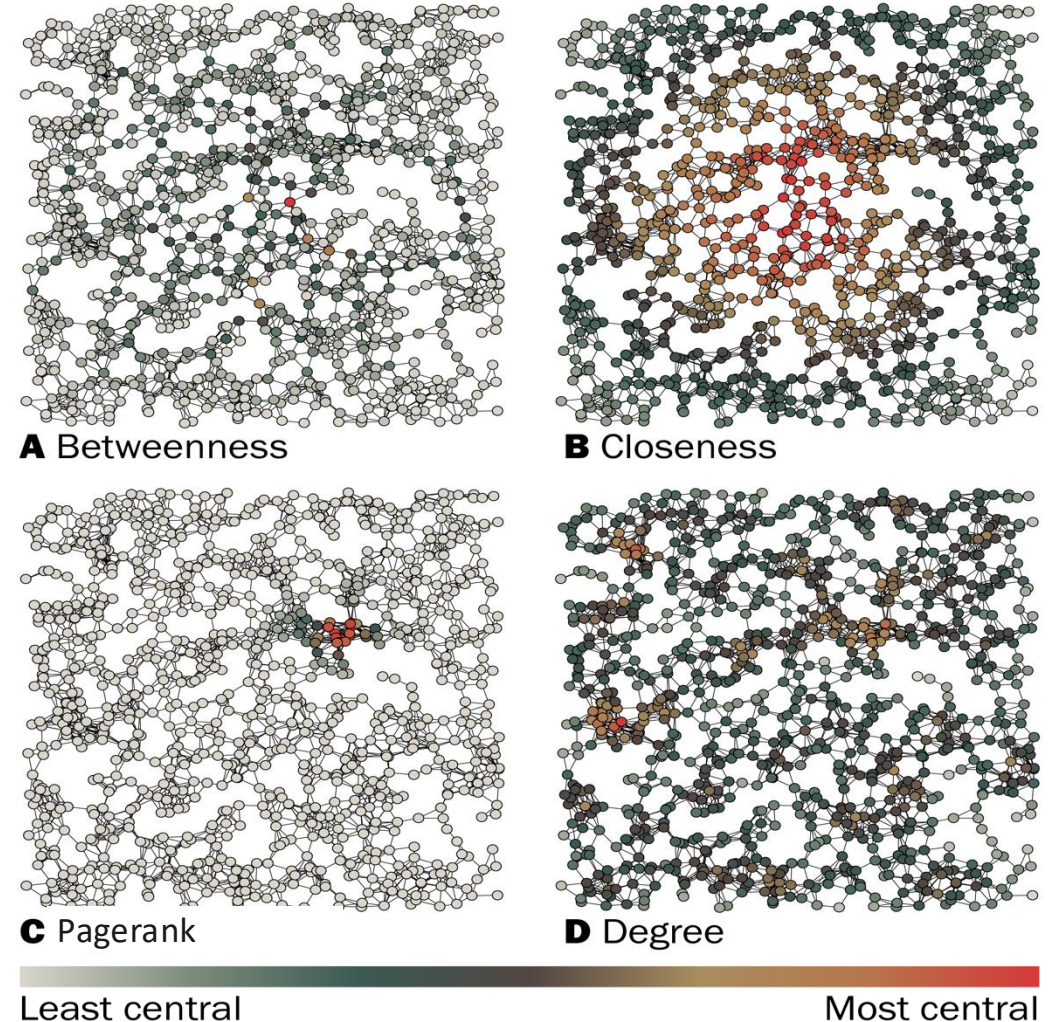
Who are the key actors in the network?

Centrality measures provide answers to this question.

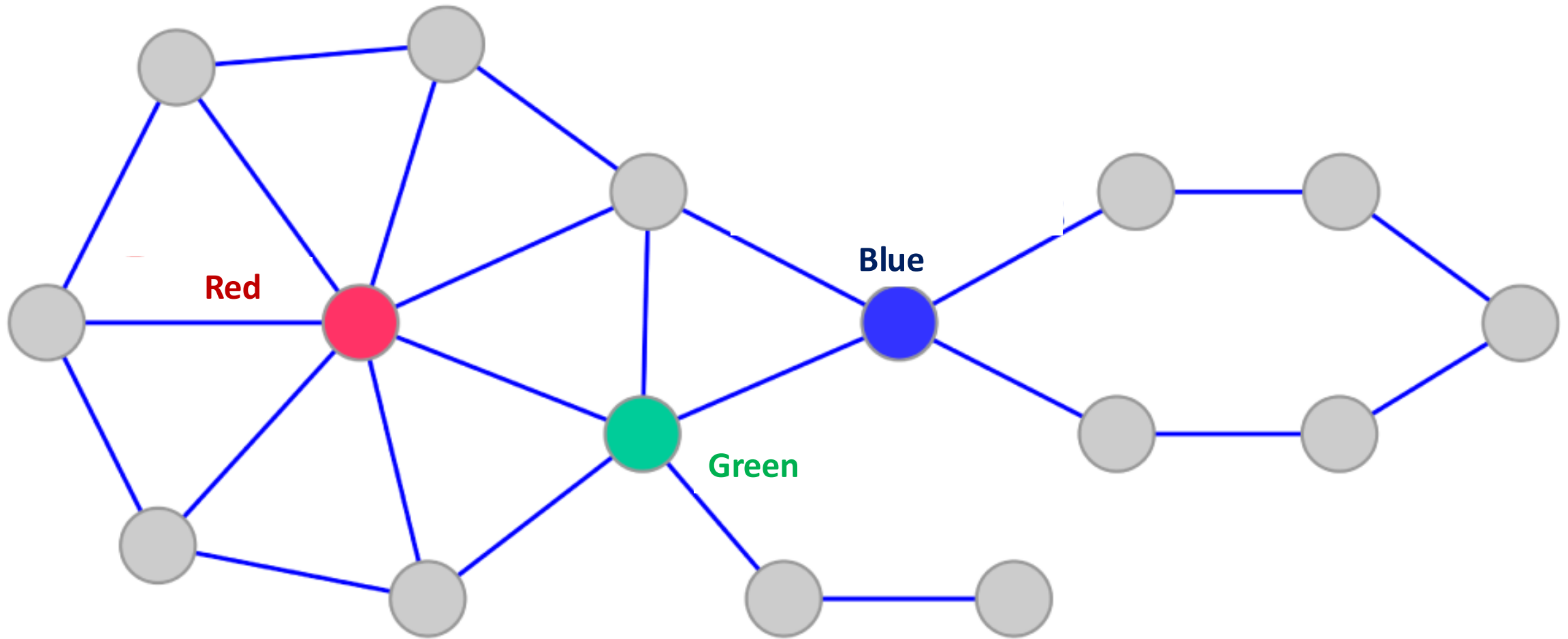
Different centrality measures define importance in different ways:

- *Degree*: Connected to many nodes
- *Closeness*: Close to all other nodes
- *Betweenness*: In the middle of shortest paths
- *Pagerank*: Connected to important nodes

Centrality identifies the most important *nodes*. It does not quantify the importance of nodes in general. The relative rankings of non-important nodes may be meaningless.



Which node has higher degree/betweenness/closeness?

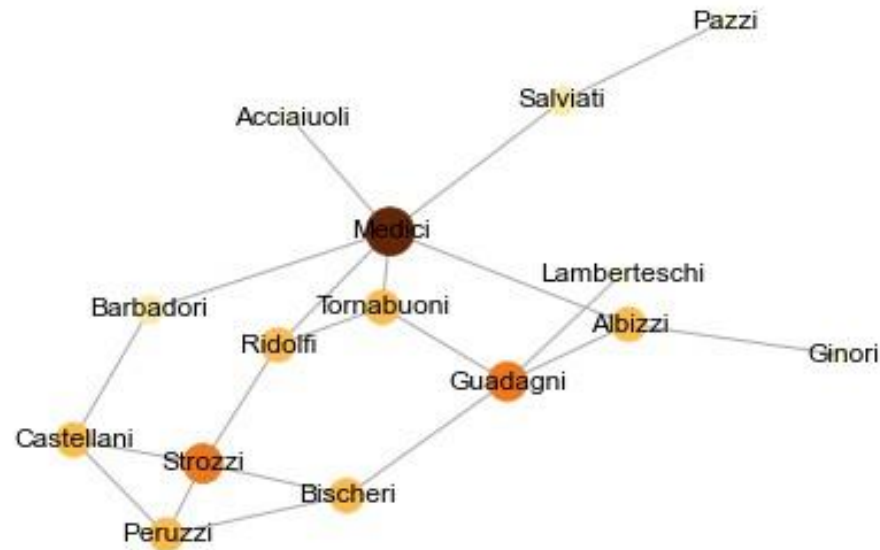


Degree centrality: $DC_i \propto d_i$

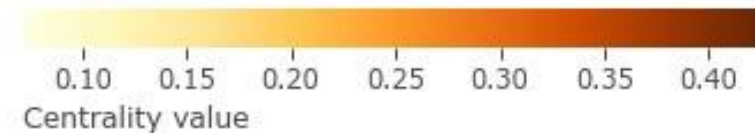
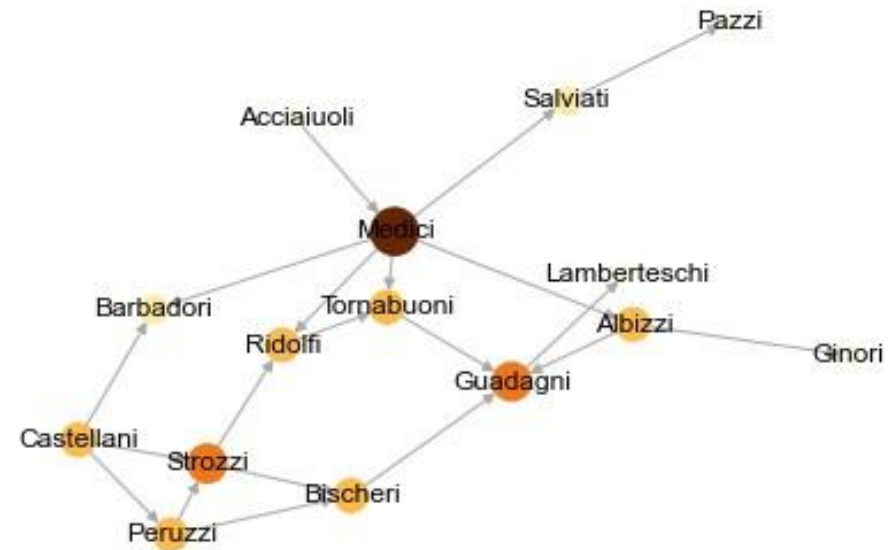
d_i = degree of node i

Measures the **local** influence of the node

Undirected



Directed



Closeness centrality: $CC_i = \frac{1}{l_i} = \frac{N}{\sum_j d_{ij}}$

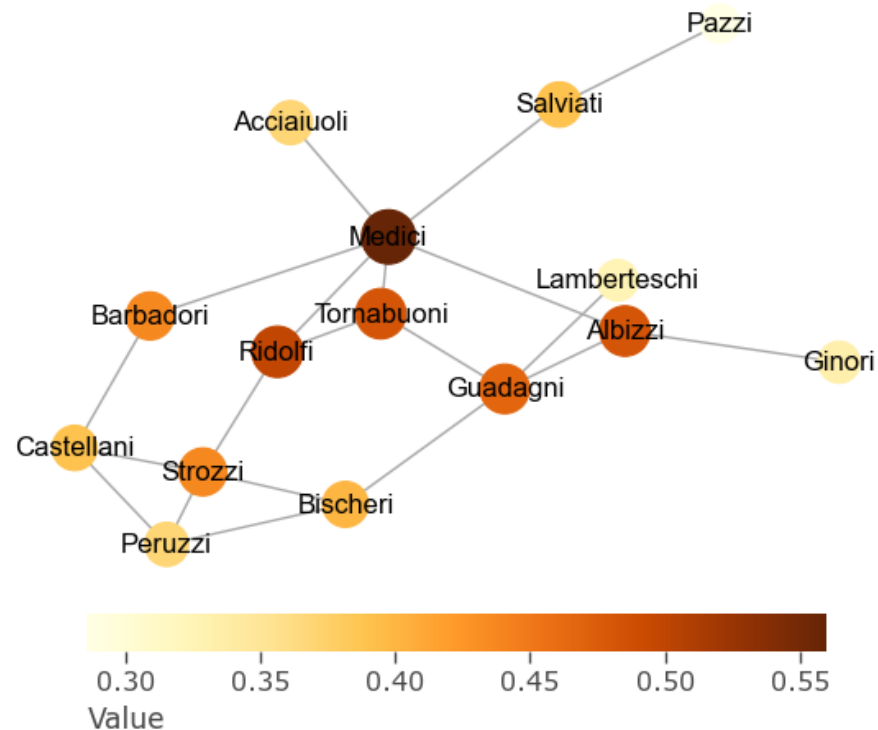
l_i = average distance of node i to all other nodes

d_{ij} = shortest distance from node i to node j

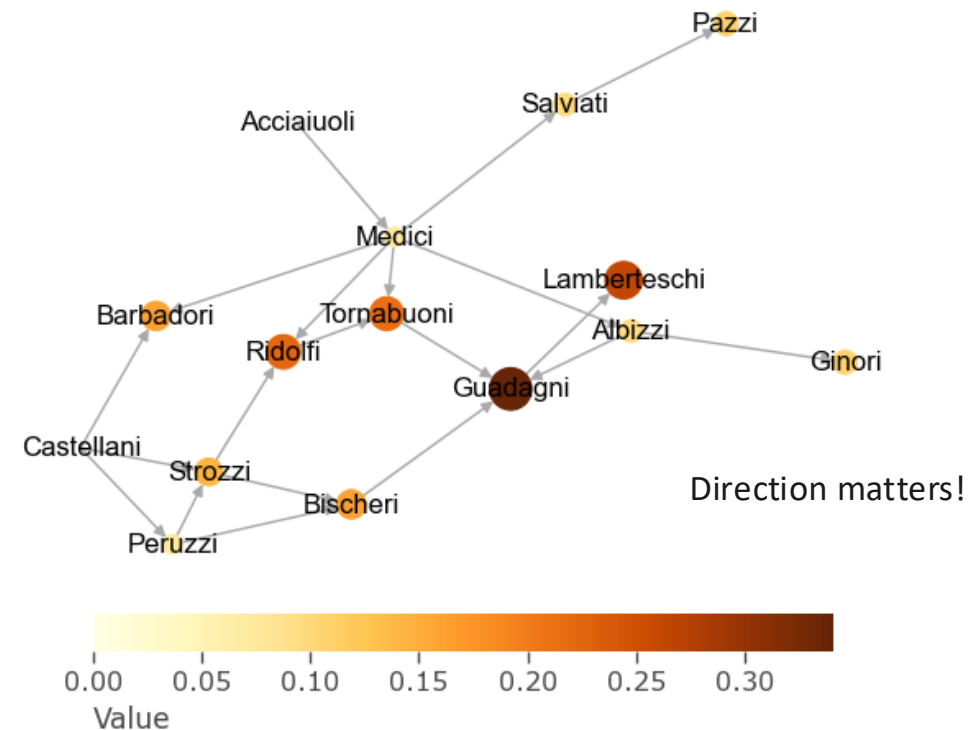
Only useful in fully connected networks

Measures the **most central** node in the network (closest to get to all other nodes)

Undirected



Directed



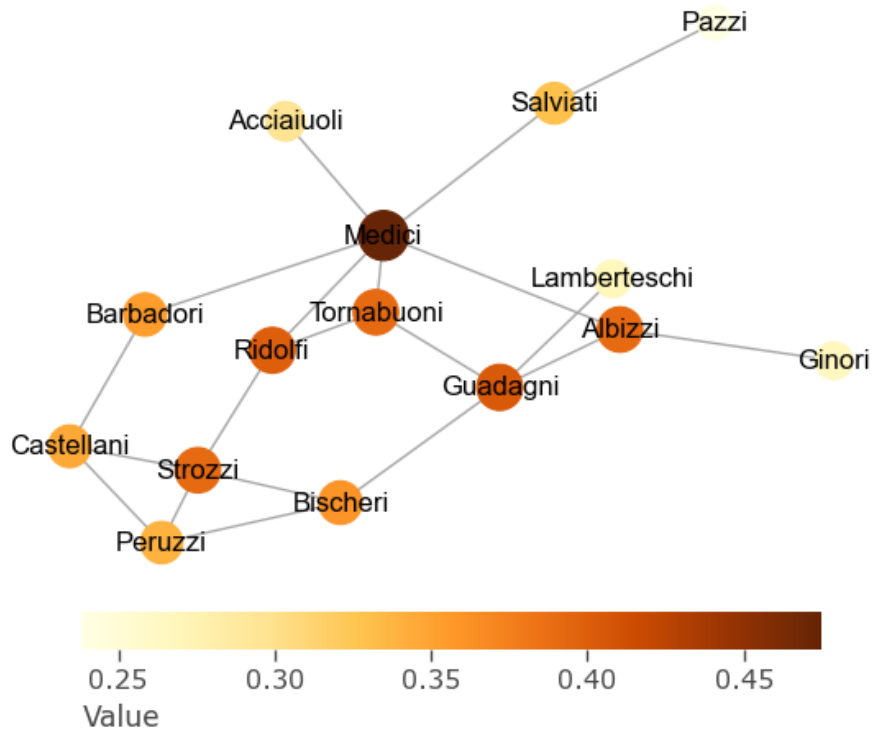
Harmonic closeness centrality: $HCC_i \propto \sum_{ij|i \neq j} 1/d_{ij}$

d_{ij} = shortest distance from node i to node j

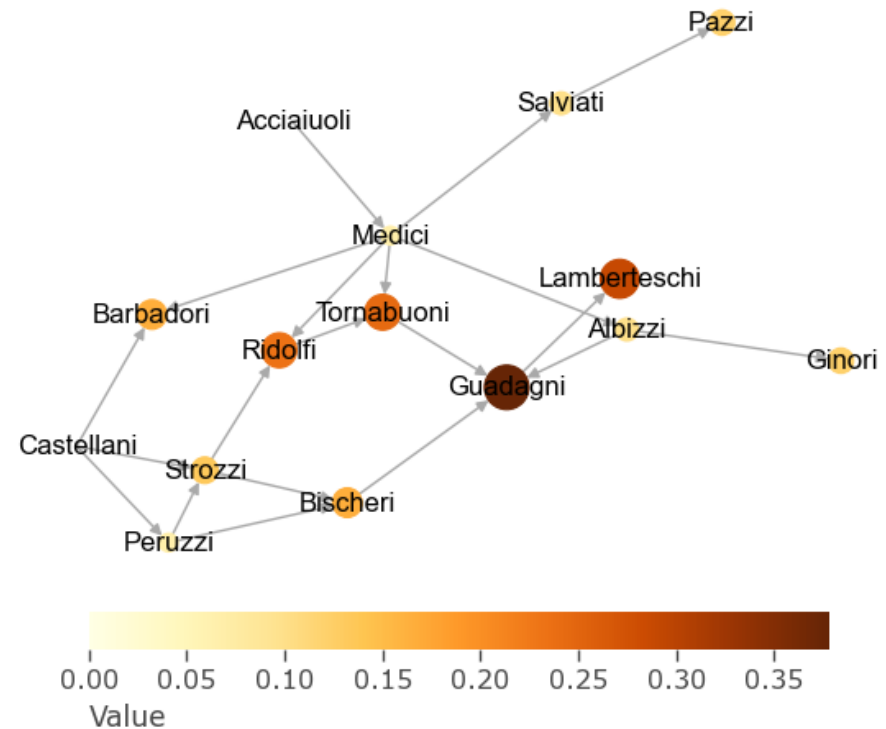
Useful also in disconnected networks. Gives more weight to closer nodes.

Measures the **most central** node in the network (harmonic average)

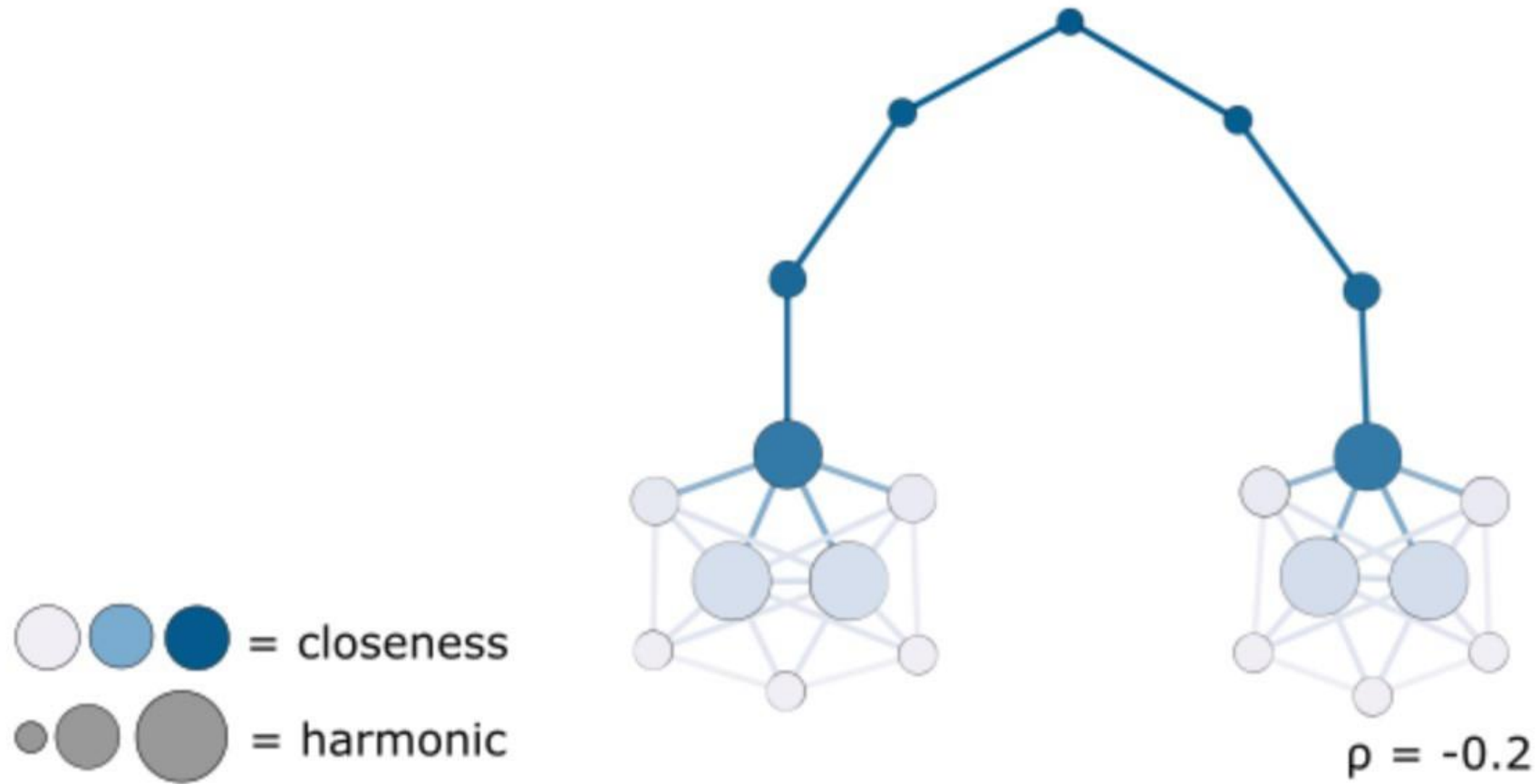
Undirected



Directed



Closeness vs harmonic closeness



Betweenness centrality: $BC_i \propto \sum_{st} n_{st}^i$

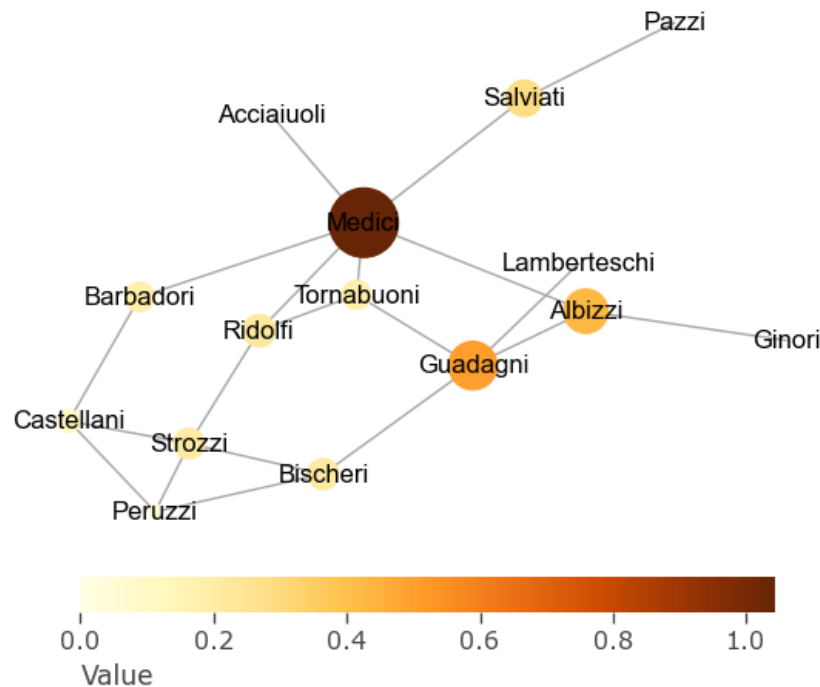
$n_{st}^i = 1$ if node i lies on the shortest path between nodes s and t (or $1/g$ if there are g shortest paths)

Measures **brokerage** in the network → disruption of these nodes = disruption of communication

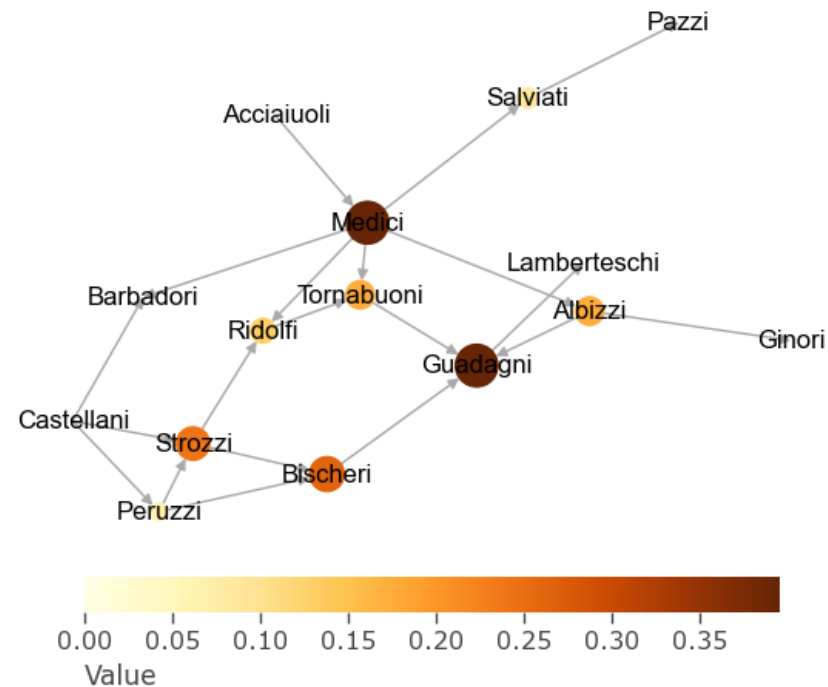
Assumptions:

- every pair of nodes in the network exchanges messages at the same average rate
- messages take the shortest available path through the network

Undirected



Directed



*Freeman (1977), and
Anthonisse (1971,
unpublished)*

Eigenvector centrality: $EC_i = \lambda^{-1} \sum_j A_{ij} EC_j$

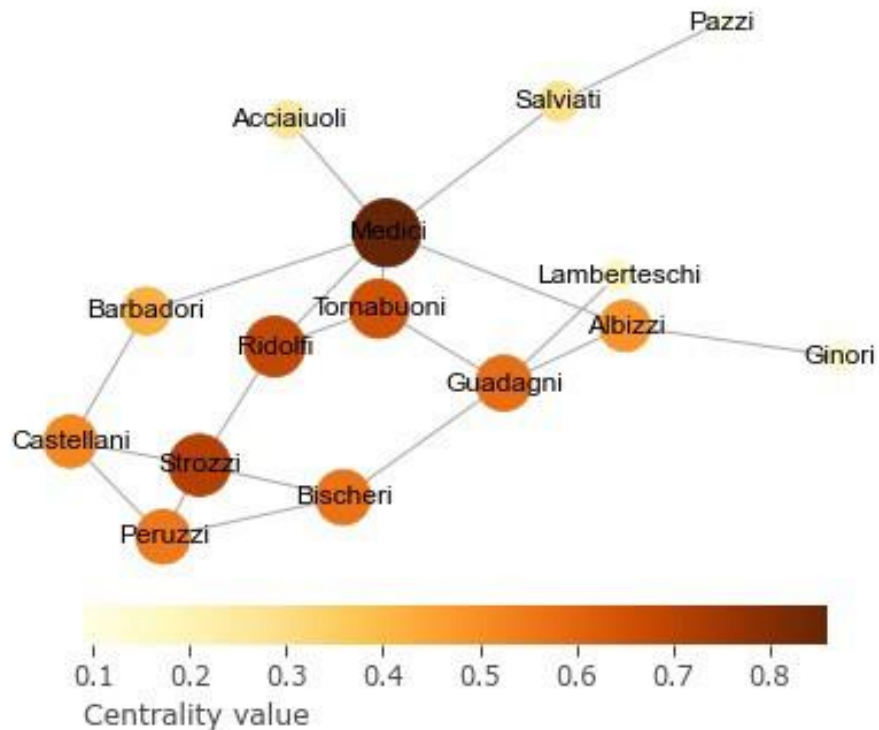
Considers how central your neighbors are.

EC_j = eigenvector centrality of node j

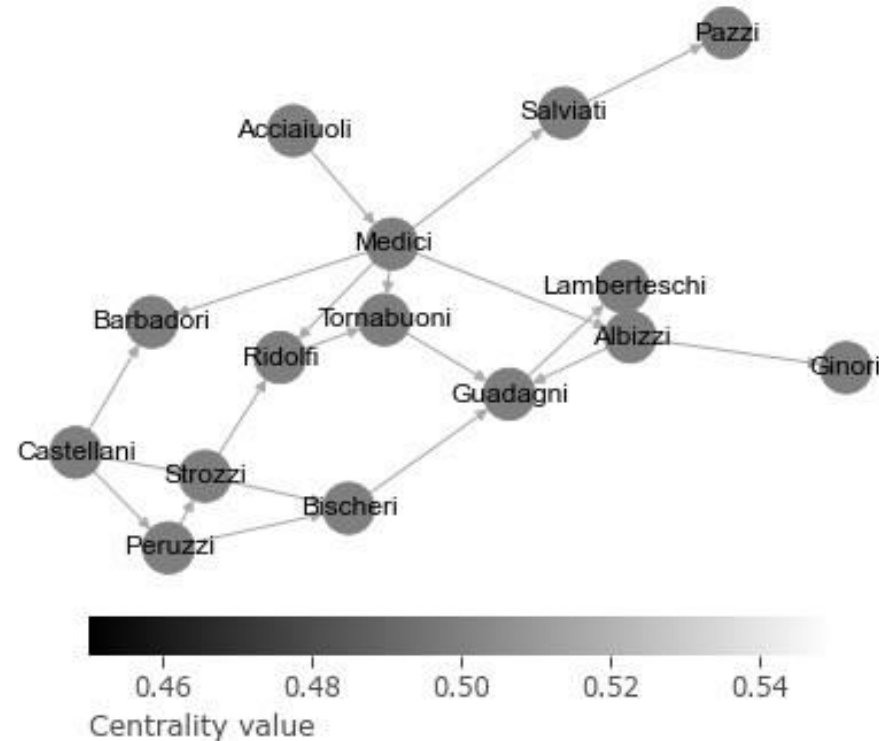
λ = largest eigenvalue

Measures total **influence** in the network (assuming all nodes are the same) Only for undirected, fully-connected networks!

Undirected



Directed



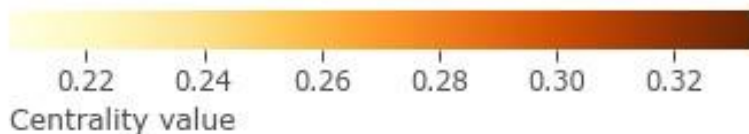
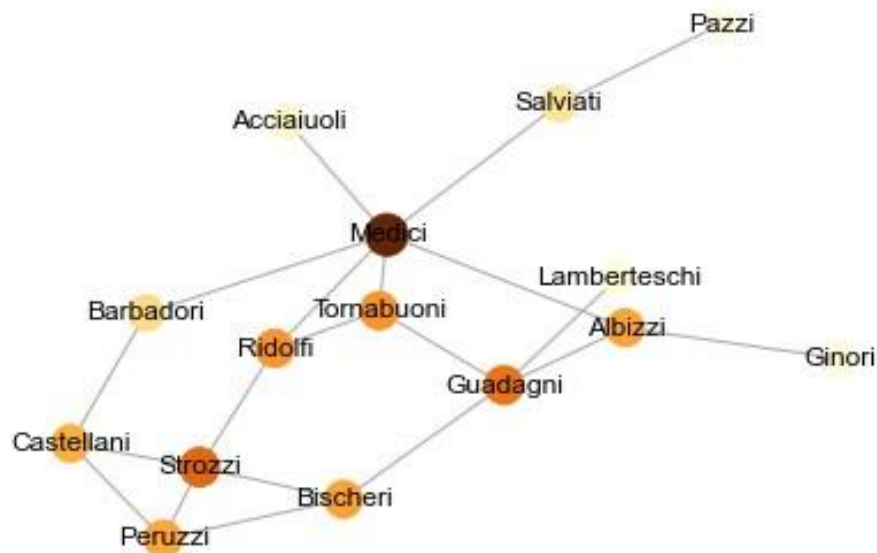
Katz centrality: $KC_i = \alpha \sum_j A_{ij} KC_j + \beta$

k_j = Katz centrality of node j

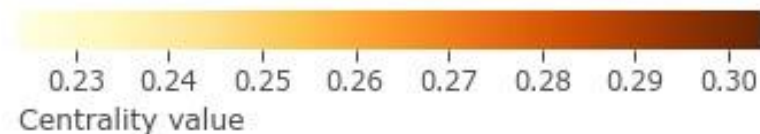
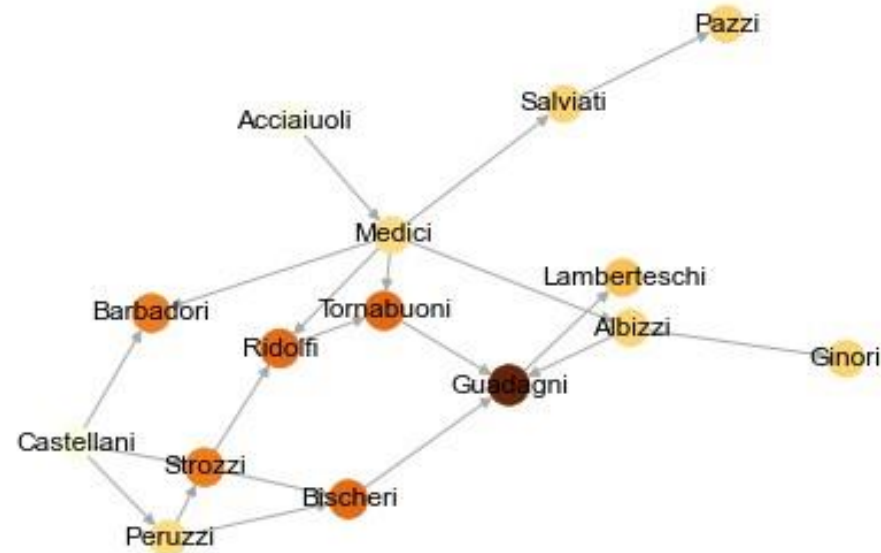
Takes into account how central your neighbors are, **each node has a minimum value of β** , and the balance between the constant and the eigenvector part is controlled by α

Measures total **influence** in the network (assuming all nodes are the same)

Undirected



Directed



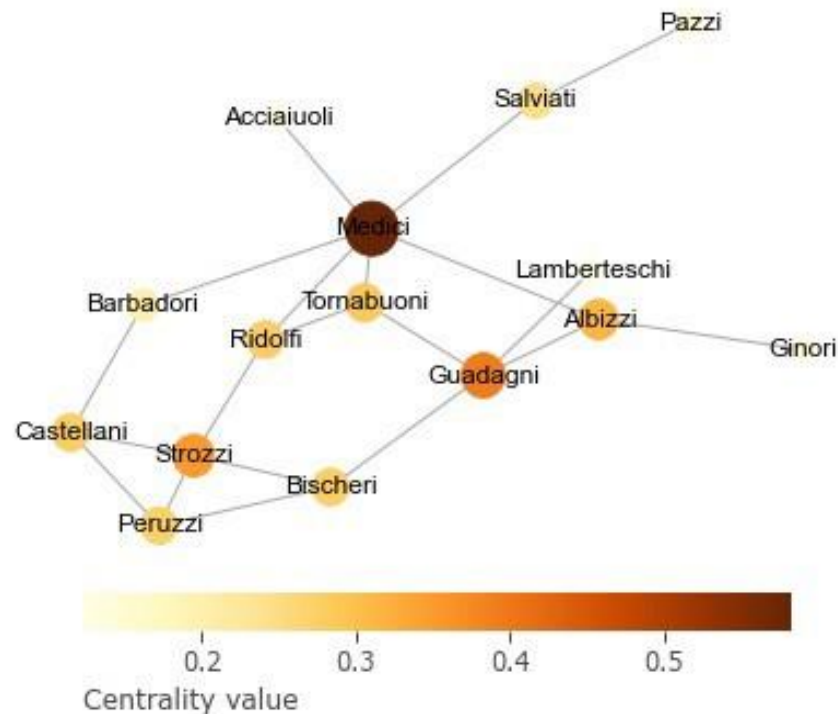
PageRank centrality: $PR_i = (1 - \alpha) \sum_j \frac{A_{ij} PR_j}{d_j} + \alpha$

d_j = Degree of node j . PR_j = Pagerank centrality of node j

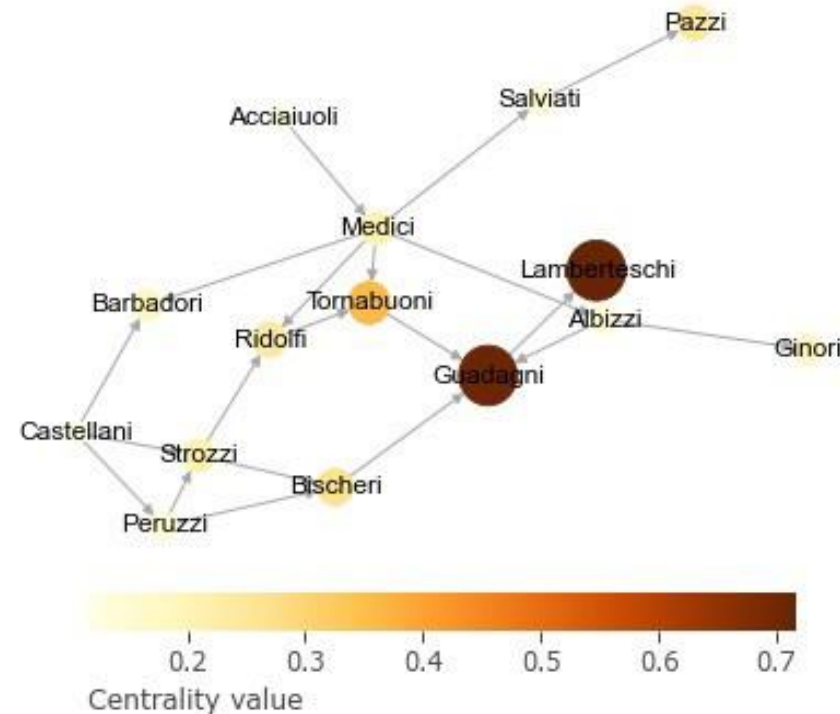
Takes into account how central your neighbors are. Each node has a minimum value of α . The PageRank of a node is α plus **the PageRank of your neighbors** (normalized by their out-degree)

Measures total **influence** in the network (assuming all nodes are similar)

Undirected



Directed



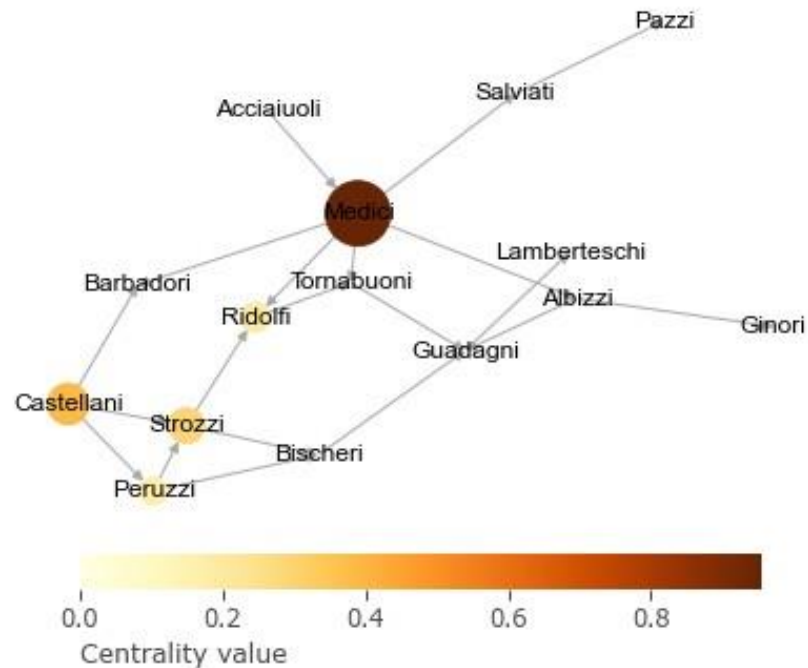
Hubs and authorities (HITS)

A node may be important if it points to others with high centrality, e.g., a review article pointing to prestigious articles

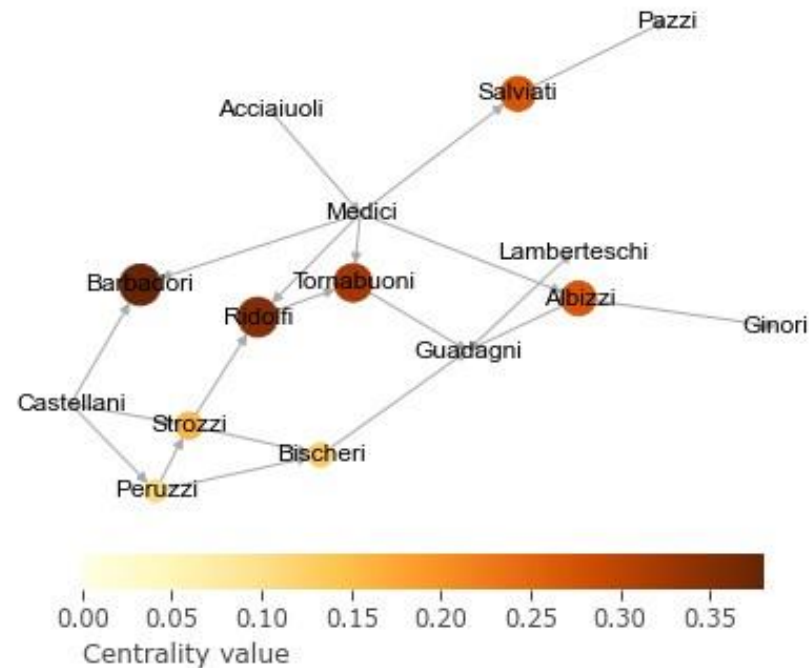
Authorities are nodes that contain useful information on a topic of interest and **hubs** are nodes that tell us where the best authorities are to be found (Newman). Two centralities: authority (a) and hub (h) centrality. **Only for directed networks!**

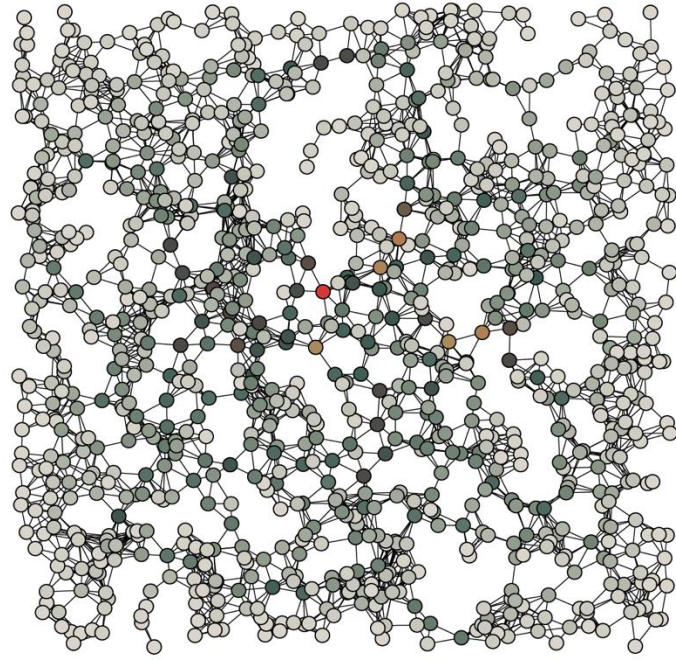
$$a_i = \alpha \sum_j A_{ij} h_j ; \quad h_i = \alpha \sum_j A_{ji} a_j$$

Hubs

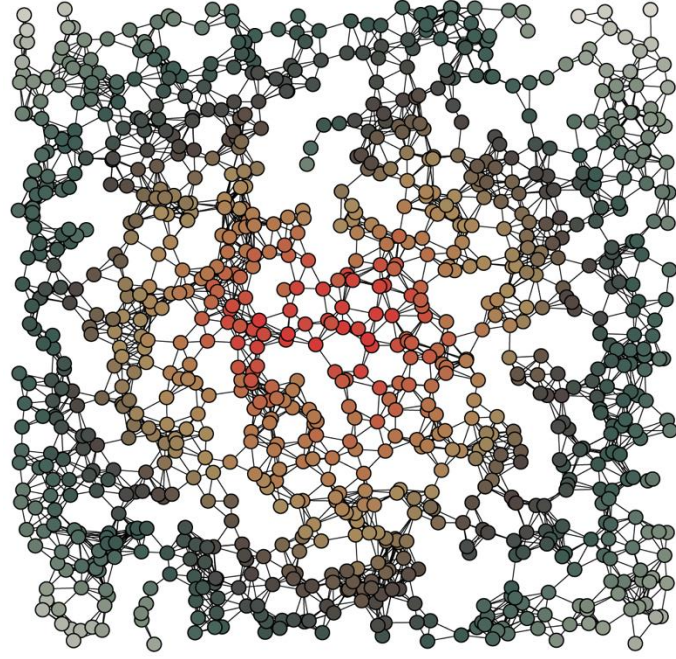


Authorities

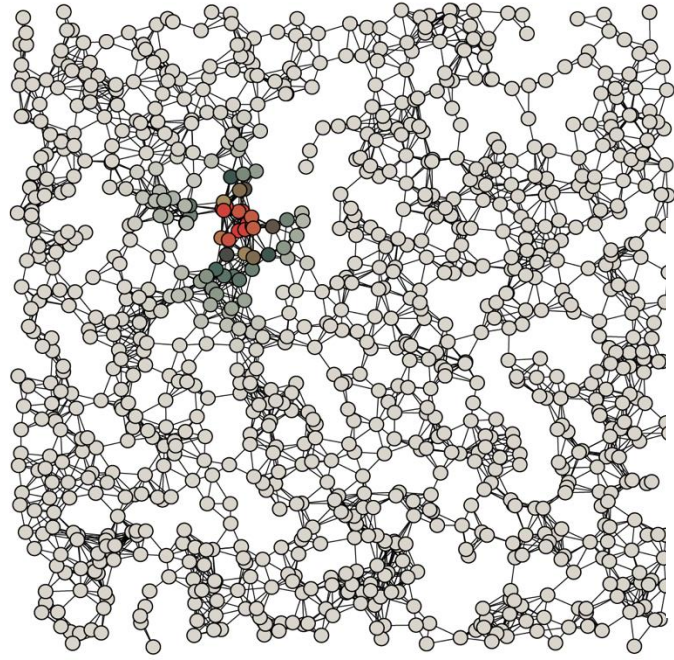




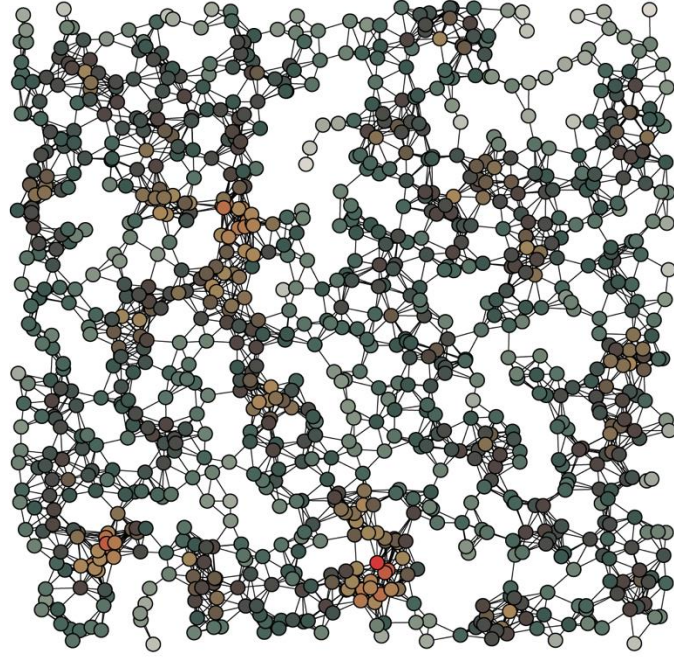
A Betweenness



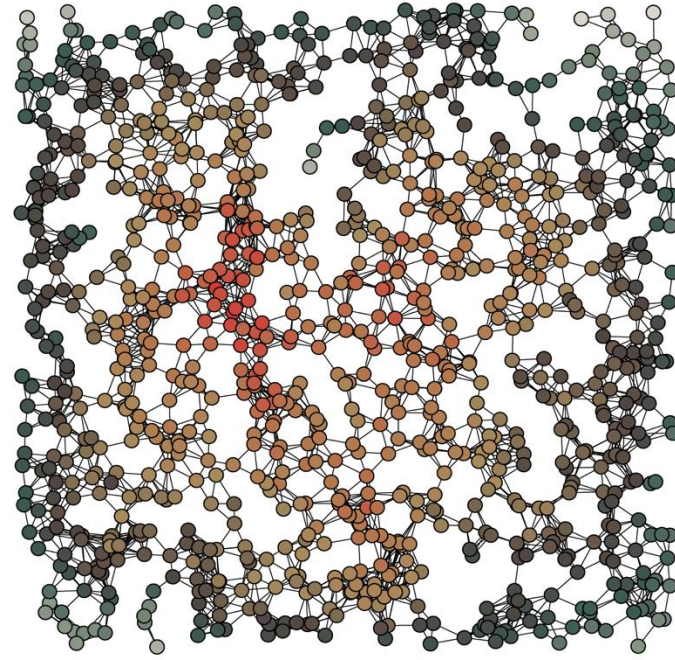
B Closeness



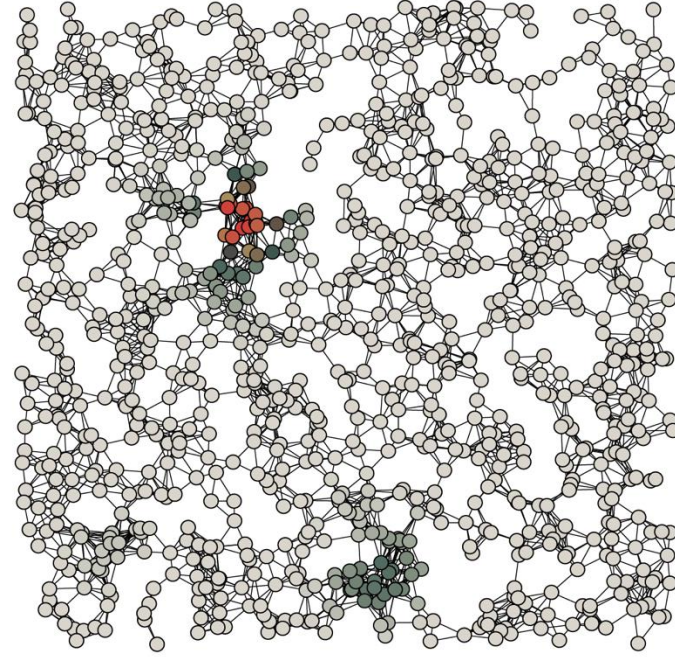
C PageRank



D Degree



E Harmonic



F Katz



Use a centrality measure that fits your theory, not the one that gives you the best results

Consider what is the real objective (e.g. is it to enable low-income individuals to increase their social capital?)
(<https://petterhol.me/2019/01/11/the-importance-of-being-earnest-about-node-importance/>)

1 IA

8000 1979

DC

Degree

2 IIA

224 1971

239 2008

BC

Betweenness

EBC

Endpoint BC

942 1966

239 2008

CC

Closeness

PBC

Proxy BC

3 IIIA

4 IVB

5 VB

6 VIB

7 VIIB

8 VIIIB

9 VIIIB

10 VIIIB

11 IB

12 IIB

13 IIIA

14 IVA

15 VA

16 VIA

17 VIIA

18 VIIIA

518 1989

IC

Information C

26 1989

275 2002

51 2004

279 1997

399 2 001

178 1995

kPC

kPath C

EGO

Ego

HYPER

Hypergraphs

AFF

Affiliation C

α -C

α -Cent

ECC

Eccentricity

9068 1999

573 2006

296 1999

80 2006

34 2010

116 1998

HITS

Hubs/Authority

g-kPC

geodesic kPath

GROUP

Groups/Cliques

HYPSC

Hypers. SC

t-SC

t-Subgraph

RAD

Radiality

1279 1972

239 2008

224 1971

53 2009

236 2007

5 2010

0 2015

2 2013

56 2007

281 1971

42 2012

427 2007

43 2009

573 2006

573 2006

505 2010

17 2013

116 1998

EC

Eigenvector

LSBC

LScaledBC

EBC

Edge BC

CBC

Commun. BC

Δ C

Delta Cent

MDC

MD Cent

EYC

Entropy C

CAC

Comm. Ability

EPTC

Entropy PC

CCoef

Clust. Coef

PeC

PeC

BN

Bottleneck

EI

Essentiality I

e-kPC

e-disjoint kPC

v-kPC

v-disjoint kPC

WEIGHT

Weighted C

TCom

Total Comm

INT

Integration

1306 1953

239 2008

979 2005

477 1991

42 2009

11 2008

0 2014

45 2012

0 2015

1 2014

4 2012

119 2008

43 2009

179 2005

426 1988

116 1991

58 2007

586 2004

KS

Katz Status

DBBC

DBounded BC

RWBC

RWalk BC

TEC

Total Effects

LI

Lobby Index

MC

Mod. Cent

COMCC

Community C

ECCoef

ECCoef

SMD

Super Mediat

UCC

United Comp

WDC

WDC

MNC

MNC

KL

Clique Level

BIP

Bipartivity

GPI

GPI Power

kRPC

Reachability

SCodd

odd Subgraph

RWCC

RWalk CC

8053 1999

239 2008

291 1953

477 1991

1 2014

10 2012

0 2012

1699 2001

0 2015

15 2011

26 2011

119 2008

3 2013

2457 1987

X X

27 2012

13 2007

0 2014

PR

Page Rank

DSBC

DScaled BC

σ

Stress

IEC

Immediate Eff

DM

Degree Mass

LAPC

Laplacian C

ABC

Attentive BC

STRC

Straightness C

SNR

Silent Node R

HPC

Harm. Prot

LAC

Local Average

DMNC

DMNC

LR

Lurker Rank

β -C

β -Cent

HYP

Hyperbolic C

kEPC

k-edge PC

FC

Functional C

HCC

Hierar. CC

484 2005

613 1991

14 2012

477 1991

69 2010

35 2010

X X

15 2010

14 2013

11 2013

45 2012

108 2010

X X

1 2014

36 2009

0 2014

0 2014

0 2015

SC

Subgraph

FBC

Flow BC

RLBC

RLimited BC

MEC

Mediative Eff

LEVC

Leverage Cent

TC

Topological C

SDC

Sphere Degree

ZC

Zonal Cent

CI

Collab. Index

CoEWC

CoEWC

NC

NC

MLC

Moduland C

RSC

Resolvent SC

SWIPD

SWIPD

XXXX

LinComb

BCPR

BCPR

TPC

Tunable PC

EDCC

Effective Dist

citations year

C

Name

8000 1979

942 1966

573 2006

1130 2005

24 2014

252 1974

6 1981

3 2012

3 2009

Freeman

Sabidussi

Borgatti/Everett

Borgatti

Boldi/Vigna

Nieminen

Kishi

Kitti

Garg

Conceptual

Axiomatic

Conceptual

Conceptual

Axiomatic

Axiomatic

Axiomatic

Axiomatic

Axiomatic

2065 1934

1546 1950

780 1948

1475 1951

297 1992

3649 2001

4167 1998

961 1993

71 2008

Moreno

Bavelas

Bavelas

Leavitt

Borgatti/Everett

Jeong et al.

Tsai/Ghoshal

Ibarra

Valente

Historic

Historic

Historic

Historic

Conceptual

Empirical

Empirical

Empirical

Empirical

“Traditional”

Betweenness-like

Friedkin Measures

Miscellaneous

Path-based

Specific Network Type

Spectral-based

Closeness-like

©David Schoch (University of Konstanz)

Chains

Sometimes data is represented as chains

- Life trajectory: Aranda → León → Vermont → Amsterdam
- Ownership chain: (right figure)

They allow you to do other analysis:

- Importance of the node based on how often it is found in between
- Importance of the node based on how many people jump to you

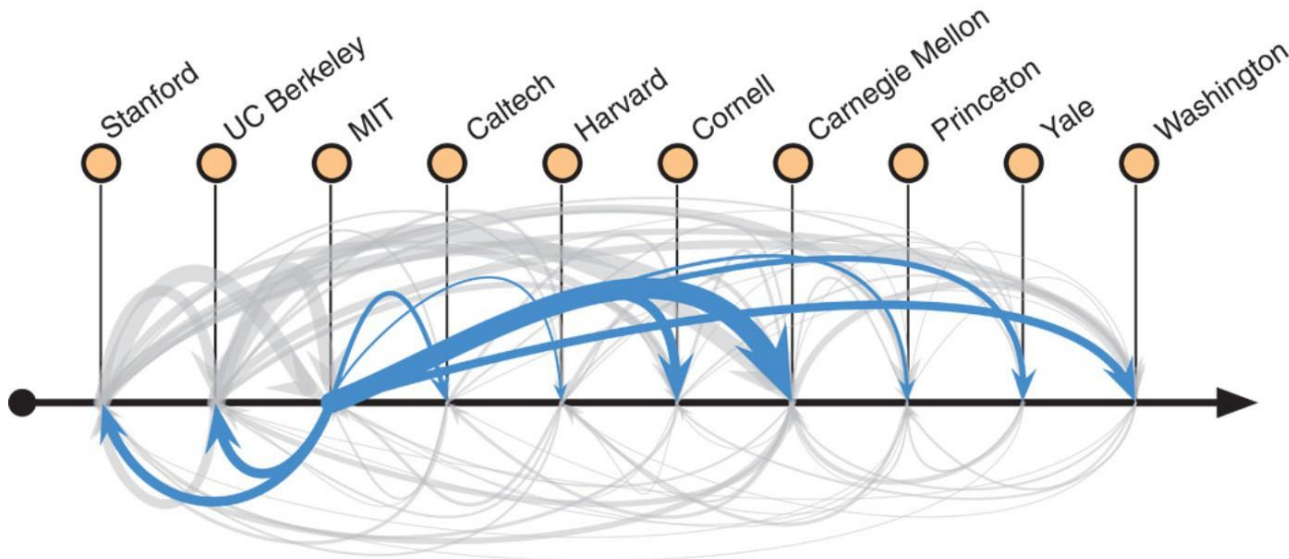


Fig. 1 Prestige hierarchies in faculty hiring networks.

Clauset, Arbesman and Larremore (2015)



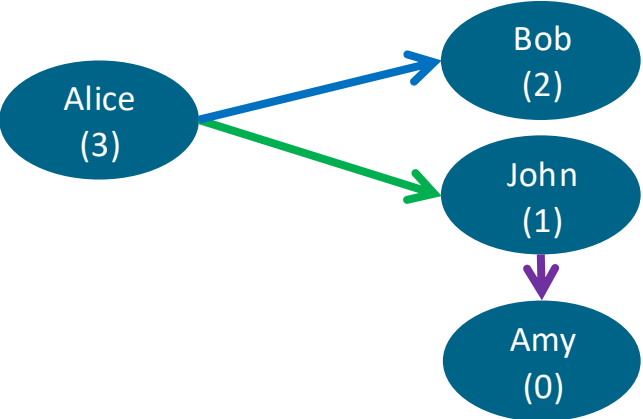
Practical 2:

Exercise 4, 5 and 7

Linear algebra and centrality measures

Matrix multiplication: paths

Interpretation A: Presence of path between node i and j



A^2

=

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=

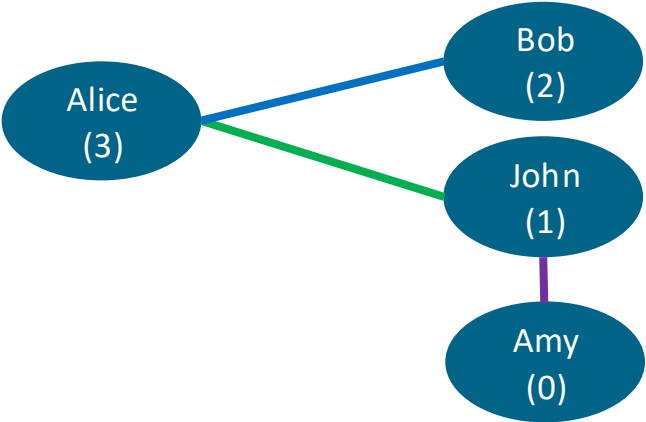
Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0	0	1
Bob	0	0	0	0
John	0	0	0	0
Amy	0	0	0	0

From you → to your neighbors

From your neighbors → to their neighbors

You → the neig. of your neig.

Degree = paths of length 1



Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Alice	1
Bob	1
John	1
Amy	1

=

	Degree
Alice	2
Bob	1
John	2
Amy	1

Eigenvector ~ paths over all possible lengths

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

@

Alice	1
Bob	1
John	1
Amy	1

=

	Degree
Alice	2
Bob	1
John	2
Amy	1

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	1	1	0
Bob	1	0	0	0
John	1	0	0	1
Amy	0	0	1	0

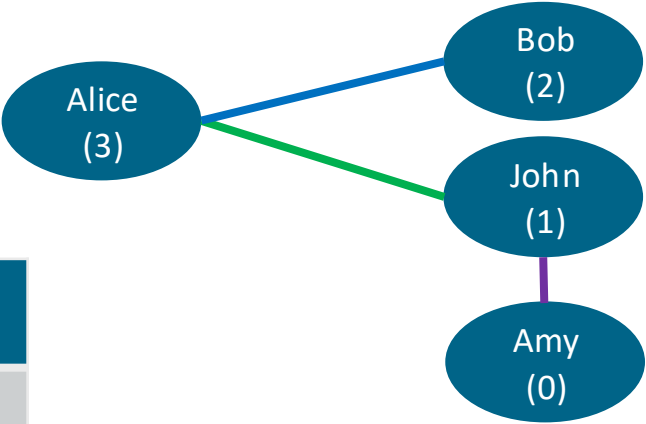
@

Alice	2
Bob	1
John	2
Amy	1

=

Alice	3
Bob	2
John	3
Amy	2

...



Another view on matrix multiplications: Random walks on undirected networks

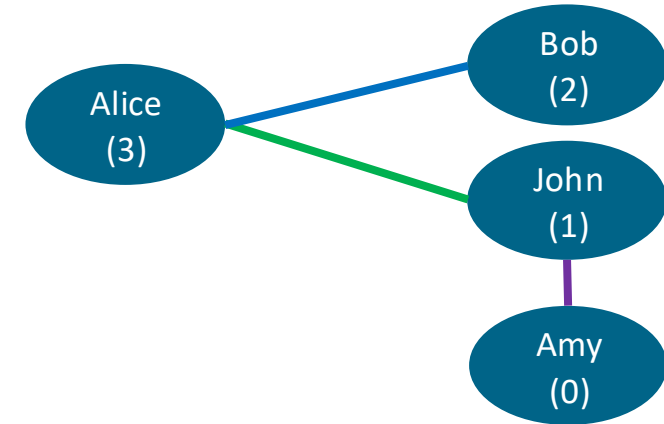
Transition matrix (row-normalized A)

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0.5	0.5	0
Bob	1	0	0	0
John	0.5	0	0	0.5
Amy	0	0	1	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0.5	0.5	0
Bob	1	0	0	0
John	0.5	0	0	0.5
Amy	0	0	1	0

=



A random walker starting in Alice will go 50% of the times to Bob, 50% of the times to John.

- From Bob it goes 100% of the times back to Alice
- From John it goes 50% of the times to John, 50% back to Alice

If we let the random walker walk forever → The fraction of time spend at each node converges to the **degree centrality** of the node

Another view on matrix multiplications: Random walks on directed networks

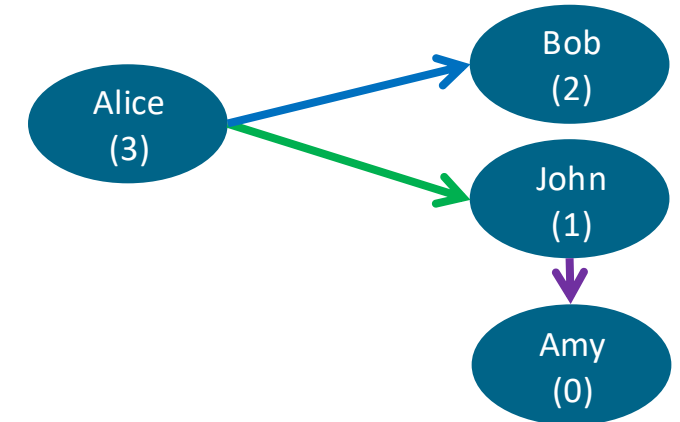
Transition matrix (row-normalized A)

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0.5	0.5	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

@

Target → ↓ Source	Alice	Bob	John	Amy
Alice	0	0.5	0.5	0
Bob	0	0	0	0
John	0	0	0	1
Amy	0	0	0	0

=



A random walker starting in Alice will go 50% of the times to Bob, 50% of the times to John.

- From Bob it gets trapped
- From John it goes 100% of the times to Amy and gets trapped

If we let random walkers walk forever → They get trapped in the extremes!

Solution: PageRank (the alpha parameter can be understood as a teleportation probability)

Practical 3:

Working with networks using Gephi

- Follow this tutorial (slides 1–23 only!): <https://gephi.org/users/quick-start/>
- In community detection use the “stochastic blockmodel” instead of modularity maximization (or try both)
- You can choose to use our own data (<https://tinyurl.com/network-game>) or the Twitter or PPI data.

Python exercise notebook 2, ex.6 (if you like math)

Recap of today

There is important information encoded in relationships/interactions

Modeling systems using networks allows us to study that information

We can represent networks using adjacency matrices or adjacency lists

Network science is closely linked to linear algebra (matrix multiplication).

You can now:

- **Describe networks:** number of edges and nodes, components, density, assortativity, clustering, diameter, degree distributions.
- **Test hypotheses using** reference models.
- Find the most important nodes using **centrality measures:**
 - Random paths:
 - Degree
 - Eigenvector / PageRank / Katz / Hubs and Authorities
 - Shortest paths
 - Closeness / Harmonic: Distance to all other nodes
 - Betweenness: Presence in shortest paths between other nodes

Tomorrow, with PiratePeel:

- Network Models: generate networks and test hypotheses
- Community Detection